Program: **Diploma in Artificial Intelligence & Machine Learning**	
Course Code: **5347**	Course Title: **Soft Computing Lab**
Semester: **5**	Credits: **1.5**
Course Category: **Programme Core**	
Periods per week: **3 (L:0 T:0 P:3)**	Periods per semester: **45**

Course Prerequisites:

Topic	Course code	Course name	Semester
Problem-Solving	2139	Problem Solving & Programming Lab	2
Artificial Neural Networks	4347	Machine Learning Lab	4

Course Outcomes:

On completion of the course, the student will be able to:

COn	**Description**	**Duration (Hours)**	**Cognitive Level**
CO1	Familiarize MATLAB	7	Applying
CO2	Implement neural network concepts using MATLAB	9	Applying
CO3	Implement fuzzy logic and fuzzy sets	12	Applying
CO4	Experiment with FIS Editor	14	Applying
	Lab Exam	3	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2					
CO2	2	2					
CO3	3	3					
CO4	3	3					

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Descriptio n	Duration (Hours)	Cognitive Level
CO1	**Familiarize MATLAB**		
M1.01	Familiarize MATLAB	**2**	Understanding
M1.02	Demonstrate MATLAB windows in detail. (command window, Graphics window, Editor window, Workspace, command history, current directory)	**2**	Understanding
M1.03	Illustrate basic MATLAB commands with example	**3**	Applying
CO2	**Implement neural network concepts using MATLAB**		
M2.01	Generate AND NOT function using McCulloch-Pitts neural net.	3	Applying
M2.02	Generate XOR function using McCulloch-Pitts neural net.	3	Applying
M2.03	Implement a feedforward neural network for a classification task	3	Applying
CO3	**Implement fuzzy logic and fuzzy sets**		
M3.01	Apply Union, Intersection and Complement operations using Fuzzy Logic	3	Applying

M3.02	Make use of fuzzy logic to implement De'Morgan's law	3	Applying
M3.03	Simulate fuzzy logic controllers	3	Applying
M3.03	Experiment with various membership functions	3	Applying
CO4	**Implement FIS Editor**		
M4.01	Implement FIS Editor. Use the Fuzzy toolbox to model the tip value given after a dinner based on quality and service.	4	Applying
M4.02	Implement FIS Editor. (Example Valve status based on Water Level in a water tank)	4	Applying
M4.03	Open-ended experiment	6	Applying
	Lab Exam – I	3	

Text / Reference

T/R	Book Title/Author
R1	S.N. Sivanandam, S.N. Deepa, " Principles of Soft Computing", Wiley India, 2nd Ed.,2011.

Online Resources

Sl.No	Website Link
1	https://www.tutorialspoint.com/matlab/matlab_commands.htm
2	https://www.mathworks.com/matlabcentral/fileexchange/2173-neuro-fuzzy-and-soft-computing